Bilal Aytekin

Ph.D. Student in Mathematics, University of Maryland

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SUMMARY

Mathematics Ph.D. student at the University of Maryland with B.Sc. degrees in both Mathematics and Computer Engineering. Research in number theory and arithmetic geometry, and co-author of MathAdv, a diagnostic benchmark for formal theorem provers. Instructor and teaching assistant for university courses in calculus, differential equations, linear algebra, and probability.

EDUCATION

University of Maryland, College Park 2023 – present
Ph.D. in Mathematics College Park, MD

- Advisor: Dori Bejleri.
- Research: higher-genus motivic height zeta functions for weighted projective stacks.

Boğaziçi University 2018 – 2023
B.Sc. in Mathematics and B.Sc. in Computer Engineering (double major) Istanbul, Turkey

- Thesis (Computer Engineering): *Entropy Hierarchy*. Advisor: Cem Say.

SKILLS

Programming: Python, C++, Java
Formal verification: Lean 4, Mathlib
Computer algebra: SageMath, Magma, PARI/GP
Typesetting: LaTeX

PUBLICATIONS AND PREPRINTS

1. **Higher-genus motivic height zeta functions for weighted projective stacks**.
   In preparation.
2. **MathAdv: What Theorem Provers Know, Reason, Formalize, and Generalize**.
   With J. Yuan, C. Martinez Lockhart, X. Liu, J. Wang, C. Deng, X. Han, V. Mastrantonis, D. Gudin, S. Zhu, A. Mohamed, J. Lang, Z. Song, and F. Huang. Submitted, 2026. arXiv:2608.25449.
3. **An Alexandrov Topology for Maximal Cohen–Macaulay Modules**.
   With M. Akdenizli, B. Çetin, and Ö. Esentepe. *Journal of Pure and Applied Algebra* 228 (2024), no. 1. arXiv:2210.03532.

TEACHING EXPERIENCE

University of Maryland, Department of Mathematics 2023 – present
Instructor; Teaching Assistant College Park, MD

- Instructor for Introduction to Math Modeling and Probability (Fall 2025).
- Teaching assistant for Calculus II (Fall 2023, Fall 2024), Calculus III (Spring 2024), Differential Equations (Spring 2025), and Linear Algebra (Spring 2026).

Boğaziçi University Fall 2022
Undergraduate Assistant; Grader Istanbul, Turkey

- Assistant for Graduate Algebra II; grader for Introduction to Probability and Statistics for Computer Engineers.

AWARDS AND HONORS

TÜBİTAK 2205 Undergraduate Scholarship 2019 – 2023
Scientific and Technological Research Council of Turkey (TÜBİTAK)